ECOMMERCE INSIGHTS & OPTIMIZATION

Final Analytics Report | Power BI Dashboard Project

Executive summary

This project analyzes ecommerce order-line records to understand sales concentration, category performance, freight-cost exposure, and time-based patterns. The Power BI report is designed to support category prioritization, shipping-efficiency investigations, and more disciplined planning. The analysis is descriptive: it identifies where to investigate and what to measure, but it does not estimate accounting profit or prove the causes of observed patterns.

ANALYSIS RECORDS	RECORDED SALES	FREIGHT CHARGES	FREIGHT / SALES
57,284	**R$ 6.88M**	**R$ 1.15M**	**16.65%**
Rows retained after period filter	Sum of price field	Sum of freight_value	Freight divided by recorded sales

What the analysis tells us

- Sales are concentrated. The top five categories contribute 39.4% of recorded sales; the top ten contribute 62.1%.
- Freight deserves focused review. Total freight equals 16.65% of recorded sales, with high category-level burdens in Electronics, Office Furniture, Furniture & Decor, and Housewares.
- Watches & Gifts leads by sales value. It records R$ 624,435.49 in sales (9.1% of total) with an aggregate freight-to-sales ratio of 8.3%.
- May 2018 is the highest observed month. It records R$ 567,036.98 based on the supplied shipping-date proxy; this is not sufficient by itself to establish recurring seasonality.

Core conclusion

The clearest near-term opportunity is to protect the categories that already drive sales while investigating shipping burden in categories where freight consumes a relatively large share of sales value. Any claimed improvement should be verified against follow-up operational and margin data.

1. Business Context & Method

Business question

Management needs a concise view of which categories and products contribute most to recorded sales value, where freight costs are comparatively burdensome, and how recorded sales vary over time. The dashboard is organized around these questions rather than around a list of unrelated visuals.

Dataset and analysis scope

Item	Observed value / treatment
Source file	dashboard_data.csv (24 columns)
Original rows	57,373
Analysis-period rows	57,284 rows where in_analysis_period = TRUE
Observed date range	08 Oct 2016 to 30 Aug 2018 in ship_date
Distinct product IDs	21,604
Category labels	74 including unmatched / unknown records
Product match status	56,464 matched; 820 unmatched (1.43%)

Analytical workflow

- 1. Filter: retained records marked **in_analysis_period = TRUE**; 89 of the original 57,373 records were excluded by the supplied flag.
- 2. Profile: inspected field types, date coverage, categories, product matching, missing weights, and record/order identifier behavior.
- 3. Aggregate: calculated recorded sales as SUM(price), freight as SUM(freight_value), and freight burden as total freight divided by total recorded sales.
- 4. Compare: ranked categories and products by sales value, reviewed category freight ratios, summarized monthly sales, and calculated rank-based associations.
- 5. Communicate: translated descriptive patterns into proposed actions with measurable KPIs and explicit limitations.

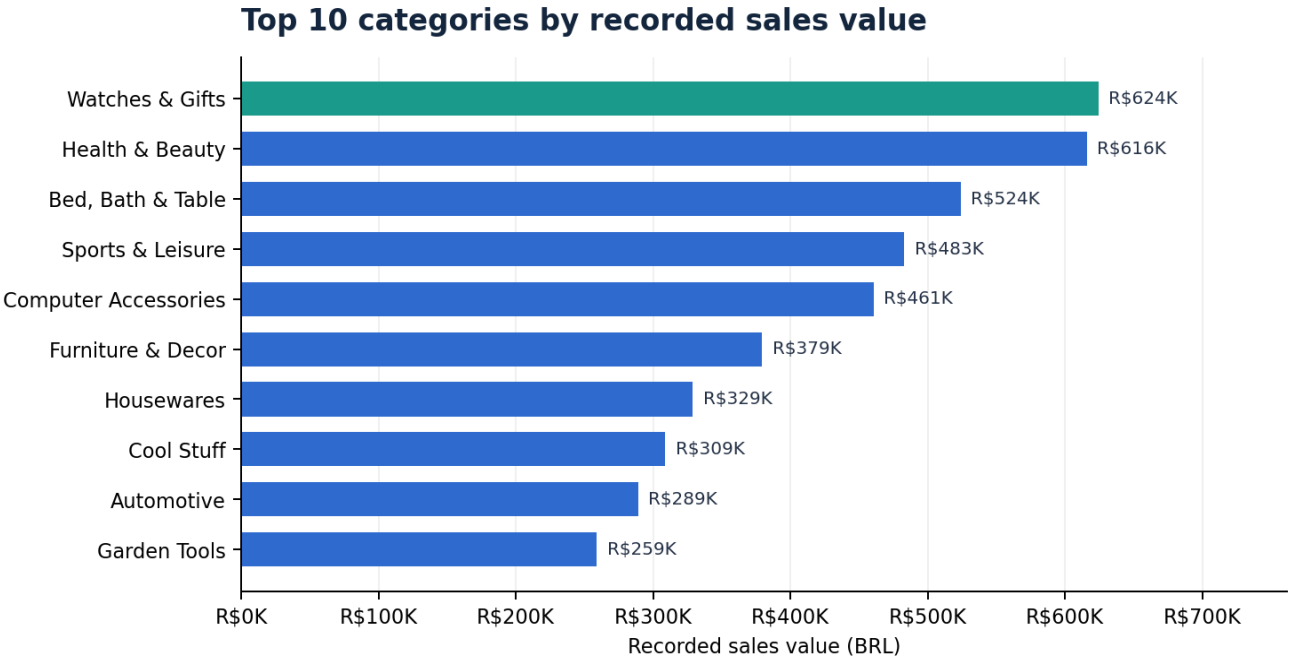
Metric definitions and interpretation

Metric	Definition used	Interpretation caution
Recorded sales value	SUM(price)	Price is treated as the sales-value field per the supplied Build Kit; validate against source business definitions.
Freight cost	SUM(freight_value)	Represents recorded freight charges, not total logistics or fulfillment cost.
Freight-to-sales ratio	SUM(freight_value) / SUM(price)	A burden indicator, not a gross-margin or profit measure.
Average value per record	Recorded sales / row count	Do not label as a validated AOV without confirming the order grain.
Monthly trend	Sales grouped by month_start / ship_date proxy	The supplied date is documented as a shipping-deadline proxy, not a confirmed order date.

Data and metric definitions follow the supplied project brief and Power BI Build Kit; any business meaning not explicit in the file is treated as an assumption, not a fact.

2. Category & Product Performance

The analysis-period extract contains 57,284 records and 74 category labels. Sales are not evenly distributed: the top five categories contribute 39.4% of total recorded sales value and the top ten contribute 62.1%. This concentration makes category-level availability and promotion decisions potentially material to overall sales performance.



Leading categories

Category	Sales value	Share	Freight	Freight / sales
Watches & Gifts	R$ 624,435.49	9.1%	R$ 51,783.26	8.3%
Health & Beauty	R$ 615,877.94	8.9%	R$ 91,128.90	14.8%
Bed, Bath & Table	R$ 524,146.52	7.6%	R$ 103,892.54	19.8%
Sports & Leisure	R$ 483,046.68	7.0%	R$ 83,893.45	17.4%
Computer Accessories	R$ 460,675.58	6.7%	R$ 75,540.20	16.4%
Furniture & Decor	R$ 379,387.42	5.5%	R$ 89,288.37	23.5%
Housewares	R$ 328,817.60	4.8%	R$ 75,480.98	23.0%
Cool Stuff	R$ 308,925.56	4.5%	R$ 41,453.37	13.4%

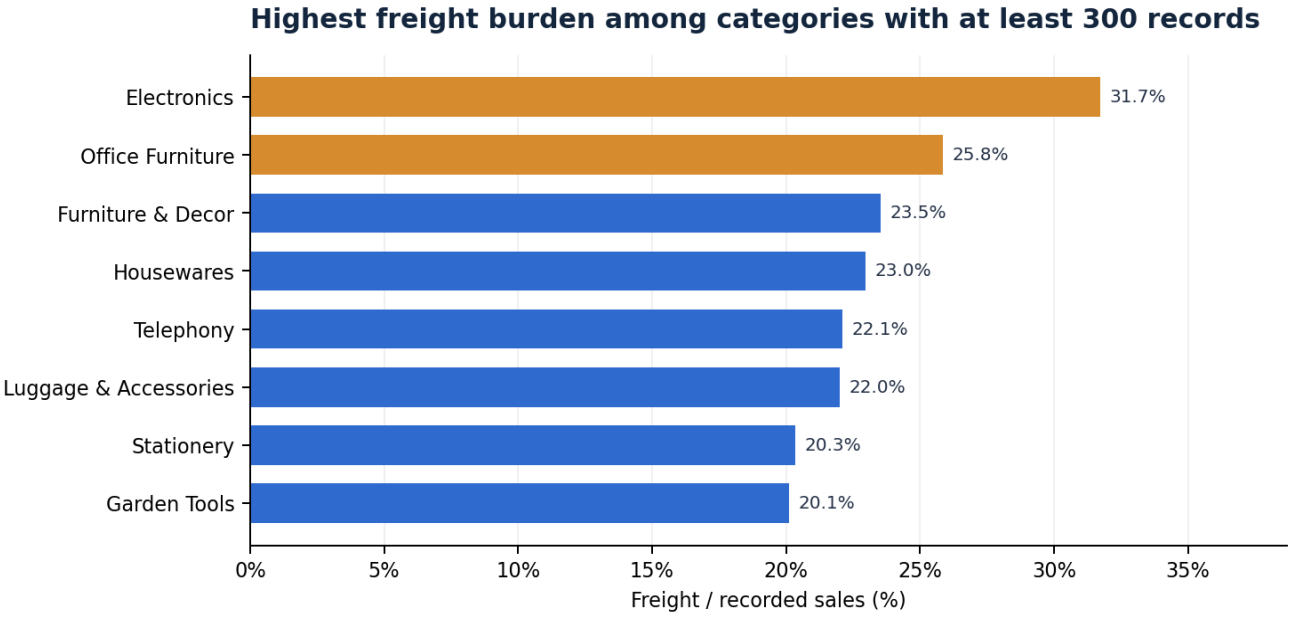
Interpretation

Watches & Gifts is the leading category by recorded sales (R$ 624,435.49, 9.1% share) and has a relatively low freight-to-sales ratio (8.3%). Health & Beauty is close behind at R$ 615,877.94. Bed, Bath & Table is a major contributor in absolute terms but carries a higher freight burden of 19.8%.

Product-level rankings are useful for identifying high-sales items, but the product ID is an identifier rather than a customer-friendly name. Before acting on a ranking, join a reliable product catalogue and consider availability, returns, margin, and sales sustainability.

3. Shipping Cost & Efficiency

Total recorded freight is R$ 1,146,145.90, equal to 16.65% of recorded sales value. Absolute freight totals alone can be misleading because high-volume categories may naturally incur more shipping charges. The freight-to-sales ratio provides a normalized view of shipping burden.



What stands out

- Electronics: 31.7% freight-to-sales across 1,406 records/orders in this extract. It is a high burden, although it contributes only 1.1% of total sales.
- Office Furniture: 25.8% freight-to-sales across 850 records/orders.
- Furniture & Decor and Housewares: burdens of 23.5% and 23.0%, respectively, are paired with meaningful sales contribution (5.5% and 4.8%). These are good candidates for operational investigation.

Product attribute associations

Comparison	Spearman rank correlation (ρ)	Meaning
Weight vs. freight	0.45	Moderate positive association in the observed records.
Price vs. freight	0.43	Moderate positive association in the observed records.

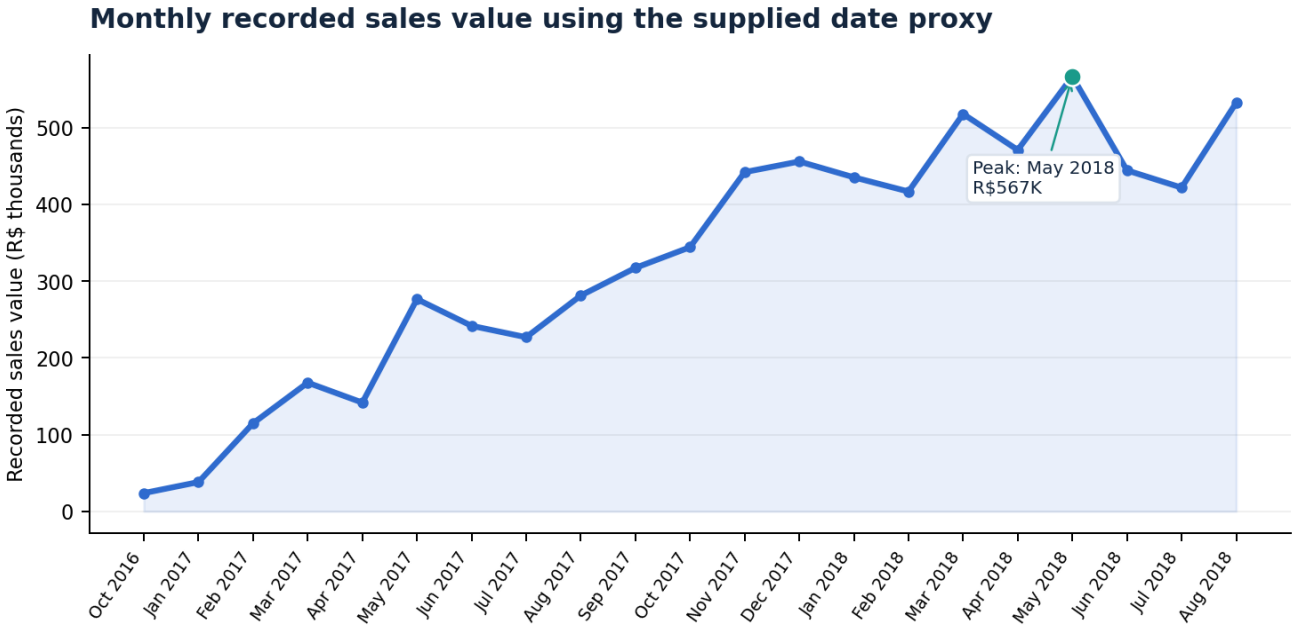
These correlations do not establish causation. Freight may also reflect dimensions, destination, carrier, delivery service, product type, and other factors not included or controlled here. The results are directional evidence for further investigation, not a model of freight pricing.

Recommended test

Select a small group of high-burden products/categories and test carrier alternatives, packaging changes, consolidation, or minimum-order thresholds. Compare freight-to-sales ratio and delivered conversion/volume before and after the change, while monitoring customer experience and total margin.

4. Time Trend & Seasonality

Recorded sales value rises over much of the observed period, with a stronger step-up in late 2017 and several high-value months in 2018. The highest observed month is May 2018 at R$ 567,036.98. August 2018 is also high at R$ 532,814.82. These observations can support planning questions, but they do not alone prove recurring seasonal behavior.



How to interpret this chart

- Observed peak: May 2018 is the highest month in the included records. Investigate campaigns, pricing, stock, marketplace events, and seller activity that may explain the peak.
- Comparisons need context: The analysis window contains no records for November or December 2016, and the supplied date is a shipping-deadline proxy. Treat month-to-month changes as patterns in the extract, not verified order-date demand.
- Seasonality is not proven: The observed period spans less than two years of recorded months, with gaps. Stronger seasonality conclusions require consistent order dates across multiple years.

Selected highest-sales months

Month	Sales value	Freight	Freight / sales
May 2018	R$ 567,036.98	R$ 87,979.92	15.5%
August 2018	R$ 532,814.82	R$ 92,609.05	17.4%
March 2018	R$ 518,182.46	R$ 91,077.31	17.6%
April 2018	R$ 471,215.49	R$ 79,213.75	16.8%
December 2017	R$ 456,171.32	R$ 73,996.03	16.2%

Planning implication

Use the peak month as a hypothesis for further investigation. Before repeating a campaign or increasing inventory ahead of the same month, confirm the date field, data completeness, and underlying drivers. Track stock-out rate, category-level sales, returns, and actual order date to distinguish true demand from operational timing.

5. Recommendations & Conclusion

Prioritized action plan

Priority	Recommendation	Suggested action	Measures to track
1	**Investigate high freight burden**	Start with Electronics, Office Furniture, Furniture & Decor, and Housewares. Review carrier/service mix, packaging dimensions, shipment consolidation, and thresholds.	Freight-to-sales %; freight per record; delivered conversion; margin after freight
2	**Protect the categories that drive sales**	The top five categories account for 39.35% of recorded sales. Review inventory cover, availability, merchandising, and promotion timing for leading categories.	Category revenue share; stock-out rate; conversion; return rate
3	**Understand the leading efficient segment**	Watches & Gifts leads by sales value with an 8.29% freight-to-sales ratio. Investigate which product or fulfillment characteristics support its comparatively low burden before assuming the pattern generalizes.	Sales value; freight-to-sales %; contribution margin when available
4	**Improve product-data completeness**	Resolve the 820 unmatched product records and review seven matched products with zero weight. Validate the product join and required attributes at data ingestion.	Unmatched-record rate; missing/zero weight count; join-match rate
5	**Validate the monthly peak before planning**	Investigate drivers of May 2018 and compare against complete order-date data across additional years. Avoid calling a single high month repeatable seasonality.	Monthly order-date sales; campaign lift; forecast error; stock-out rate

Limitations and assumptions

- Revenue is not profit. Cost of goods sold, discounts, returns, taxes, payment fees and full fulfillment costs are not present. Freight-to-sales is only a cost-pressure indicator.
- Date proxy. The project Build Kit documents ship_date as a proxy derived from a shipping deadline, not a confirmed order date. Seasonal conclusions must remain tentative.
- Order/line-item grain. All 57,284 filtered records have distinct order_id values; therefore, row count and distinct order count are both 57,284. The dataset cannot independently demonstrate multi-line order behavior, and no quantity field is provided.
- Product matches. 820 rows (1.43%) are unmatched to product attributes; 820 records have missing weight and 7 records have zero weight. Weight-based findings should account for this.
- Correlation is not causation. The relationships between weight/price and freight may reflect multiple interacting factors; they do not show that changing one field will automatically reduce freight.

Final conclusion

The project demonstrates a practical analytics workflow: prepare and validate an ecommerce extract, define interpretable metrics, compare category performance, investigate freight burden, examine time-based patterns, and translate findings into measurable actions. The most defensible conclusions are that recorded sales are concentrated in a limited set of categories and that freight burden differs materially across categories. The best next step is a controlled operational review of high-burden categories, paired with better product-data quality and more reliable order-date, quantity, and margin data. This would allow the business to move from descriptive insights toward measurable profitability improvement.

Prepared from the supplied Ecommerce Insights and Optimization project brief, Power BI Build Kit, and dashboard_data.csv. Metrics in this report were recalculated from rows where in_analysis_period = TRUE.